MSG No. 132.458 P_I4_2/mcm [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `4'22'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,4,5,6,18,19,23,24]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,7,8,17,20,21,22]`
3	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[9,12,13,14,26,27,31,32]`
4	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[10,11,15,16,25,28,29,30]`

Table 2: Wyckoff site: `4b`, site symmetry: `-42m`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,4,5,6,10,11,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,7,8,9,12,13,14]`
3	$\left[0, 0, \frac{3}{4}\right]$	`[17,20,21,22,26,27,31,32]`
4	$\left[0, 0, \frac{1}{4}\right]$	`[18,19,23,24,25,28,29,30]`

Table 3: Wyckoff site: `4c`, site symmetry: `4'/m..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,6,9,14,18,19,26,27]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,10,11,17,22,25,30]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[4,5,12,13,23,24,31,32]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[7,8,15,16,20,21,28,29]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.mm`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,6,7,8,9,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,4,5,10,11,12,13]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[17,22,23,24,25,30,31,32]`
4	[0, 0, 0]	`[18,19,20,21,26,27,28,29]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'/m`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,15,23,25]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,13,21,26]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,12,20,27]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,11,19,28]`
5	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5,10,18,29]`
6	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,16,24,30]`
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,9,17,31]`

8	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[8,14,22,32]

Table 6: Wyckoff site: 8f, site symmetry: 4'..

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	[1,6,18,19]
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[2,3,17,22]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[4,5,23,24]
4	$\left[0,\ \frac{1}{2},\ -z\right]$	[7,8,20,21]
5	$\left[\frac{1}{2},\ 0,\ -z\right]$	[9,14,26,27]
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,11,25,30]
7	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[12,13,31,32]
8	$\left[0,\ \frac{1}{2},\ z\right]$	[15,16,28,29]

Table 7: Wyckoff site: 8g, site symmetry: 2.mm

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[1,6,15,16]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[2,3,12,13]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[4,5,10,11]
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[7,8,9,14]
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[17,22,31,32]
6	$[0,\ 0,\ z]$	[18,19,28,29]
7	$[0,\ 0,\ -z]$	[20,21,26,27]
8	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[23,24,25,30]

Table 8: Wyckoff site: 8h, site symmetry: m.2m

No.	position	mapping
1	$[x,\ x,\ 0]$	[1,7,14,16]
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	[2,5,11,12]
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	[3,4,10,13]
4	$[-x,\ -x,\ 0]$	[6,8,9,15]
5	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[17,23,30,32]
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	[18,21,27,28]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	[19,20,26,29]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[22,24,25,31]

Table 9: Wyckoff site: 16i, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,23]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[2,21]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[3,20]

continued ...

Table 9

No.	position	mapping
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[4,19]
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[5,18]
6	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,24]
7	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,17]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[8,22]
9	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[9,31]
10	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[10,29]
11	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[11,28]
12	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[12,27]
13	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[13,26]
14	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[14,32]
15	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[15,25]
16	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[16,30]

Table 10: Wyckoff site: 16j, site symmetry: .2.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,4]
2	$\left[0,\ x,\ \frac{3}{4}\right]$	[2,7]
3	$\left[0,\ -x,\ \frac{3}{4}\right]$	[3,8]
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[5,6]
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[9,12]
6	$\left[0,\ -x,\ \frac{1}{4}\right]$	[10,15]
7	$\left[0,\ x,\ \frac{1}{4}\right]$	[11,16]
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	[13,14]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[17,20]
10	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[18,23]
11	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[19,24]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	[21,22]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	[25,28]
14	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[26,31]
15	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[27,32]
16	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[29,30]

Table 11: Wyckoff site: 16k, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	[2,11]
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	[3,10]
4	$\left[x,\ -y,\ \frac{1}{2}\right]$	[4,13]
5	$\left[-x,\ y,\ \frac{1}{2}\right]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]
7	$[y,\ x,\ 0]$	[7,16]

continued ...

Table 11

No.	position	mapping
8	$[-y, -x, 0]$	`[8,15]`
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$	`[17,30]`
10	$\left[\frac{1}{2}-y, x+\frac{1}{2}, 0\right]$	`[18,27]`
11	$\left[y+\frac{1}{2}, \frac{1}{2}-x, 0\right]$	`[19,26]`
12	$\left[x+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	`[20,29]`
13	$\left[\frac{1}{2}-x, y+\frac{1}{2}, 0\right]$	`[21,28]`
14	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}\right]$	`[22,25]`
15	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	`[23,32]`
16	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}\right]$	`[24,31]`

Table 12: Wyckoff site: `16l`, site symmetry: `..m`

No.	position	mapping
1	$[x, x, z]$	`[1,16]`
2	$\left[-x, x, z+\frac{1}{2}\right]$	`[2,12]`
3	$\left[x, -x, z+\frac{1}{2}\right]$	`[3,13]`
4	$\left[x, -x, \frac{1}{2}-z\right]$	`[4,10]`
5	$\left[-x, x, \frac{1}{2}-z\right]$	`[5,11]`
6	$[-x, -x, z]$	`[6,15]`
7	$[x, x, -z]$	`[7,14]`
8	$[-x, -x, -z]$	`[8,9]`
9	$\left[x+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$	`[17,32]`
10	$\left[\frac{1}{2}-x, x+\frac{1}{2}, z\right]$	`[18,28]`
11	$\left[x+\frac{1}{2}, \frac{1}{2}-x, z\right]$	`[19,29]`
12	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -z\right]$	`[20,26]`
13	$\left[\frac{1}{2}-x, x+\frac{1}{2}, -z\right]$	`[21,27]`
14	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z+\frac{1}{2}\right]$	`[22,31]`
15	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$	`[23,30]`
16	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[24,25]`

Table 13: Wyckoff site: `32m`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$\left[-y, x, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y, -x, z+\frac{1}{2}\right]$	`[3]`
4	$\left[x, -y, \frac{1}{2}-z\right]$	`[4]`
5	$\left[-x, y, \frac{1}{2}-z\right]$	`[5]`
6	$[-x, -y, z]$	`[6]`
7	$[y, x, -z]$	`[7]`
8	$[-y, -x, -z]$	`[8]`
9	$[-x, -y, -z]$	`[9]`
10	$\left[y, -x, \frac{1}{2}-z\right]$	`[10]`
11	$\left[-y, x, \frac{1}{2}-z\right]$	`[11]`

continued ...

Table 13

No.	position	mapping
12	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$[y,\, x,\, z]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z\right]$	[19]
20	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[20]
21	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[23]
24	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[28]
29	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[32]